Branch Network Design with Centralized Routing, DHCP, SNMP Monitoring, DNS & Web Services

CENG 3007, COMPUTER NETWORK

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1. INTRODUCTION

This project designs a simulated enterprise network connecting a Headquarters and two branch offices using Cisco Packet Tracer. The primary objective is to implement centralized network services, including DHCP, DNS, and HTTP, hosted on a single HQ server. Additionally, the network utilizes Static Routing for end-to-end connectivity and SNMPv2 for remote device monitoring. This report outlines the configuration steps and verification results for the proposed network topology.

2.NETWORK TOPOLOGY

The proposed network architecture utilizes a hub-and-spoke (star) topology, positioning the Headquarters Router as the central communication node. Both remote sites, Branch A and Branch B, establish connectivity to the HQ via dedicated Point-to-Point Serial WAN links. To centralize management, the HQ LAN hosts a single robust server that provides essential network services—specifically DHCP, DNS, and HTTP—for all clients across the enterprise.

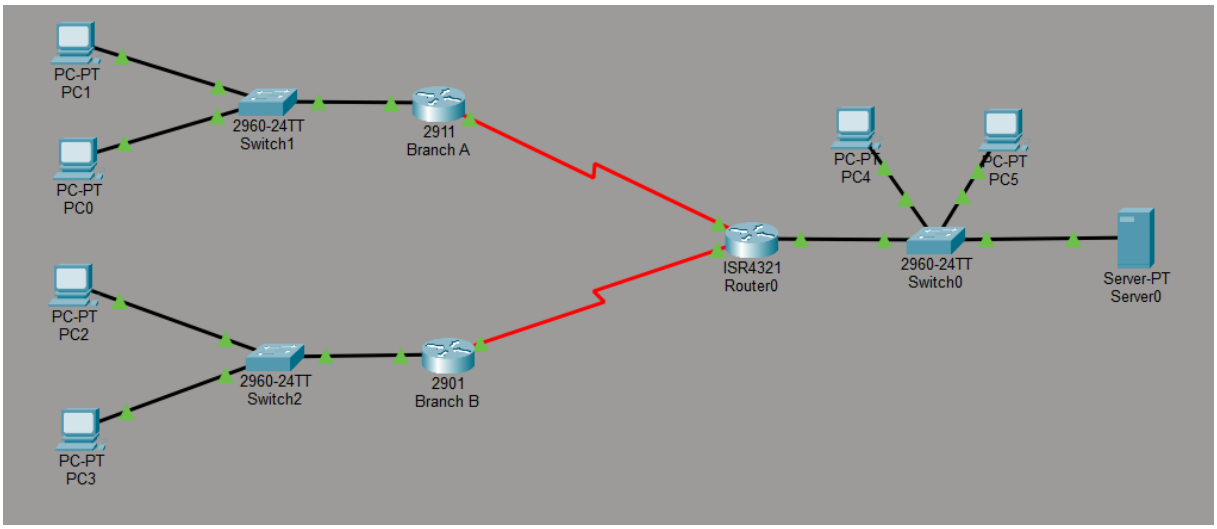


Figure 1:Network Topology Diagram

3.IP ADDRESSING TABLE

The network uses a hierarchical IP addressing scheme as defined in the requirements. The following tables detail the configuration.

Device	Interface	IP Address	Subnet Mask	Nots
HQ Router	G0/0 (LAN)	10.1.10.1	255.255.255.0	LAN Gateway
	S0/0/0 (Wan A)	192.168.100.1	255.255.255.252	Link to Branch A
	S0/0/1 (Wan B)	192.168.100.5	255.255.255.252	Link to Branch B
HQ Switch	VLAN1	10.1.10.2	255.255.255.0	Management IP
HQ Server	NIC	10.1.10.11	255.255.255.0	DHCP/DNS/Web
Branch A Router	G0/0/0(LAN)	172.16.1.1	255.255.255.0	LAN Gateway
	S0/0/0(WAN)	192.168.100.2	255.255.255.252	Link to HQ
Branch A switch	VLAN 1	172.16.1.2	255.255.255.0	Management IP
Branch B Router	G0/0/0(LAN)	172.16.2.1	255.255.255.0	LAN Gateway
	S0/0/0(WAN)	192.168.100.6	255.255.255.252	Link to HQ
Branch B Switch	VLAN 1	172.16.2.2	255.255.255.0	Management IP

Pool Name	*Network Address*	*Default Gateway*	*DNS Server*	*Address Range (Start – End)*
serverPool	*10.1.10.0 /24*	*10.1.10.1*	*10.1.10.11*	*10.1.10.21 - 10.1.10.200*
BranchA	*172.16.1.0 /24*	*172.16.1.1*		*172.16.1.10 - 172.16.1.254**
BranchB	*172.16.2.0 /24*	*172.16.2.1*		*172.16.2.10 - 172.16.2.254**

4. ROUTING CONFIGURATION

*To establish end-to-end network communication between the central site and remote locations, a **Static Routing** strategy was deployed.*

- ***Headquarters (HQ) Router:*** *Specific static route entries were added to the routing table to reach the remote subnets of **Branch A (172.16.1.0)** and **Branch B (172.16.2.0)**. These routes direct traffic through the appropriate next-hop Serial WAN interfaces.*

- ***Branch Routers:*** *To simplify the routing process at the edge, a **Default Route** (ip route 0.0.0.0 0.0.0.0) was configured on both Branch routers. This configuration forwards all outbound traffic not destined for the local LAN towards the HQ Router, acting as a gateway of last resort.*

5. SERVICE CONFIGURATION

5.1. DHCP CONFIGURATION

Network addressing is managed via a centralized DHCP architecture hosted on the HQ Server. To accommodate the specific subnet requirements of the entire enterprise, three distinct DHCP address pools were defined: one for the local HQ LAN and two for the remote Branch LANs.

To enable dynamic addressing for remote clients across the WAN, the Branch routers were configured to function as DHCP Relay Agents. This was implemented by applying the ip helper-address command to the LAN interfaces of both Branch routers, allowing them to forward local DHCP broadcast requests directly to the central HQ Server.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
BranchB	172.16.2.1	10.1.10.11	172.16.2.10	255.255.255.0	200	0.0.0.0	0.0.0.0
BranchA	172.16.1.1	10.1.10.11	172.16.1.10	255.255.255.0	200	0.0.0.0	0.0.0.0
serverPool	10.1.10.1	10.1.10.11	10.1.10.21	255.255.255.0	200	0.0.0.0	0.0.0.0

Figure 2:Centralized DHCP Pool Configuration

5.2. DNS AND WEB SERVER CONFIGURATION

The HQ Server acts as both a Web and DNS server .

- ***DNS:*** *The DNS service is enabled with an A-Record mapping www.firm.com to the Server's IP address (10.1.10.11).*
- ***Web (HTTP):*** *The HTTP service is enabled, and the index.html file was modified to display the required welcome message* ***("Welcome to Firm.com")****.*

No.	Name	Type	Detail
0	www.firm.com	A Record	10.1.10.11

Figure 3: DNS Service Configuration

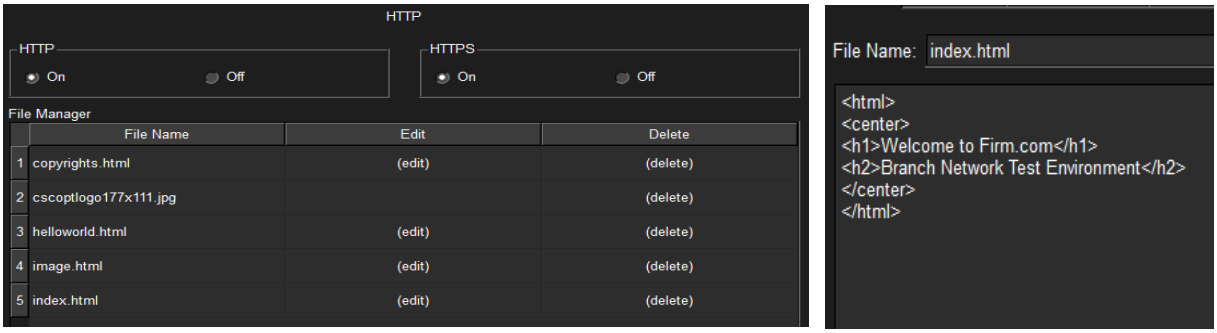


Figure 4-5 :Web Service Configuration

5.3. SNMP CONFIGURATION

Remote monitoring capabilities were implemented by configuring SNMPv2c across all network routers. The read-only community string was assigned the value 'public' to permit access to the MIB data. Additionally, the required system information, including contact and location details, was populated on every router.

6. VERIFICATION AND TESTING RESULTS

6.1. DHCP VERIFICATION

All client PCs successfully obtained an IP address, Subnet Mask, Default Gateway, and DNS Server IP automatically from the HQ Server.

```
Command Prompt                                                    X

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0060.70E6.0C42
   Link-local IPv6 Address.........: FE80::260:70FF:FEE6:C42
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     172.16.1.1
   DHCP Servers....................: 10.1.10.11
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-25-A3-E7-E9-00-60-70-E6-0C-42
   DNS Servers.....................: ::
                                     10.1.10.11
```

Figure 6:Successful DHCP Assignment on PC0 (Branch A)

```
Command Prompt                                                    X

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.9746.BBCD
   Link-local IPv6 Address.........: FE80::201:97FF:FE46:BBCD
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.2.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     172.16.2.1
   DHCP Servers....................: 10.1.10.11
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-99-00-31-0B-00-01-97-46-BB-CD
   DNS Servers.....................: ::
                                     10.1.10.11
```

Figure 7: Successful DHCP Assignment on PC3(Branch B)

6.2. CONNECTIVITY TESTS

Connectivity across the infrastructure was confirmed using standard diagnostic tools. Both Ping and Traceroute commands demonstrated successful packet transmission from the Branch networks to the Headquarters Server.

```
Pinging 10.1.10.11 with 32 bytes of data:

Reply from 10.1.10.11: bytes=32 time=5ms TTL=126
Reply from 10.1.10.11: bytes=32 time=4ms TTL=126
Reply from 10.1.10.11: bytes=32 time=3ms TTL=126
Reply from 10.1.10.11: bytes=32 time=3ms TTL=126

Ping statistics for 10.1.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 5ms, Average = 3ms
```

```
Pinging 10.1.10.11 with 32 bytes of data:

Reply from 10.1.10.11: bytes=32 time=1ms TTL=126
Reply from 10.1.10.11: bytes=32 time=5ms TTL=126
Reply from 10.1.10.11: bytes=32 time=1ms TTL=126
Reply from 10.1.10.11: bytes=32 time=7ms TTL=126

Ping statistics for 10.1.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 3ms
```

Figure 8:Ping connectivity from PC0-PC3(Branch An and BranchB) to HQ Server

6.3. WEB AND DNS VERIFICATION

Clients are able to resolve the domain name and access the corporate website.

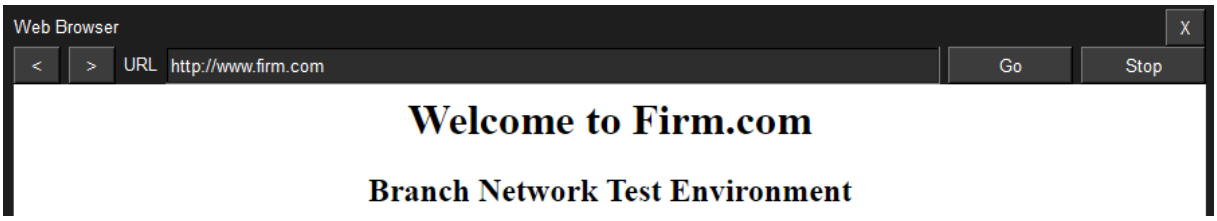
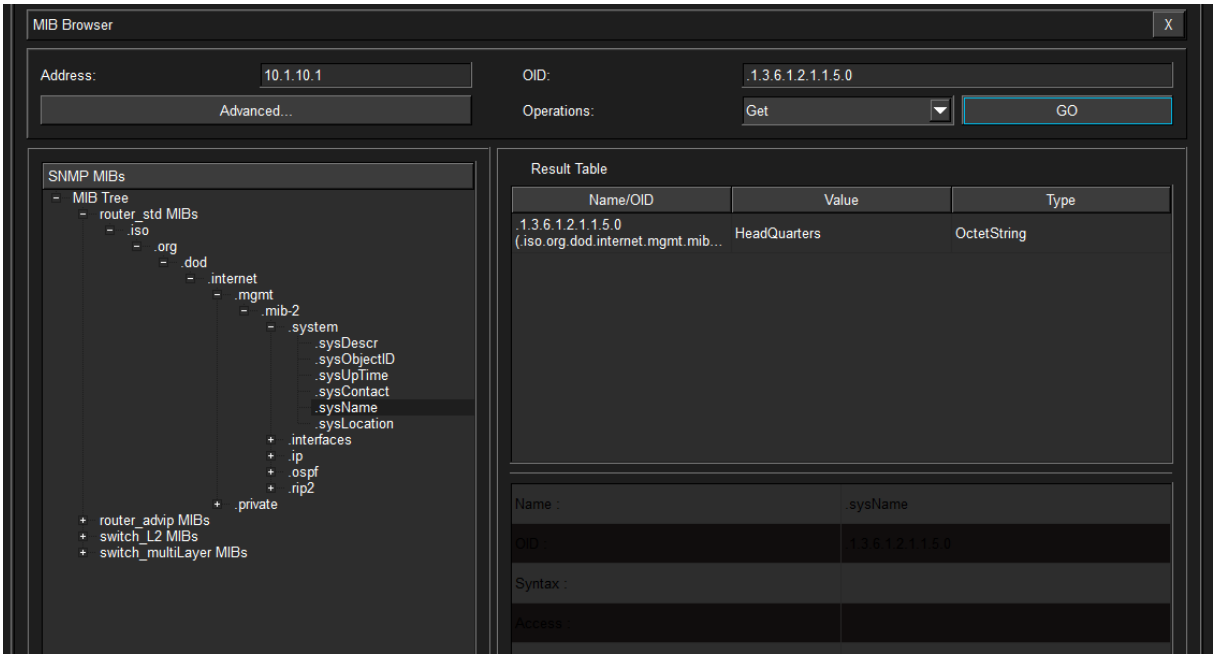
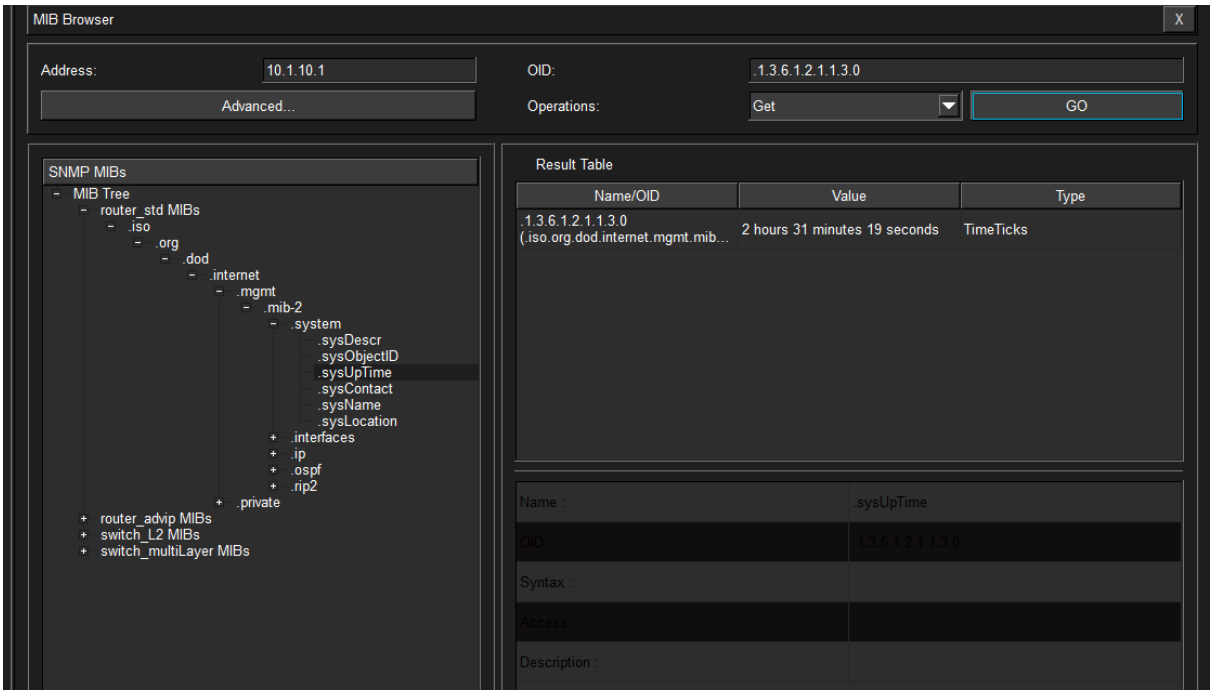


Figure 10:Succesful Web Access

6.4. SNMP MONITORING VERIFICATION





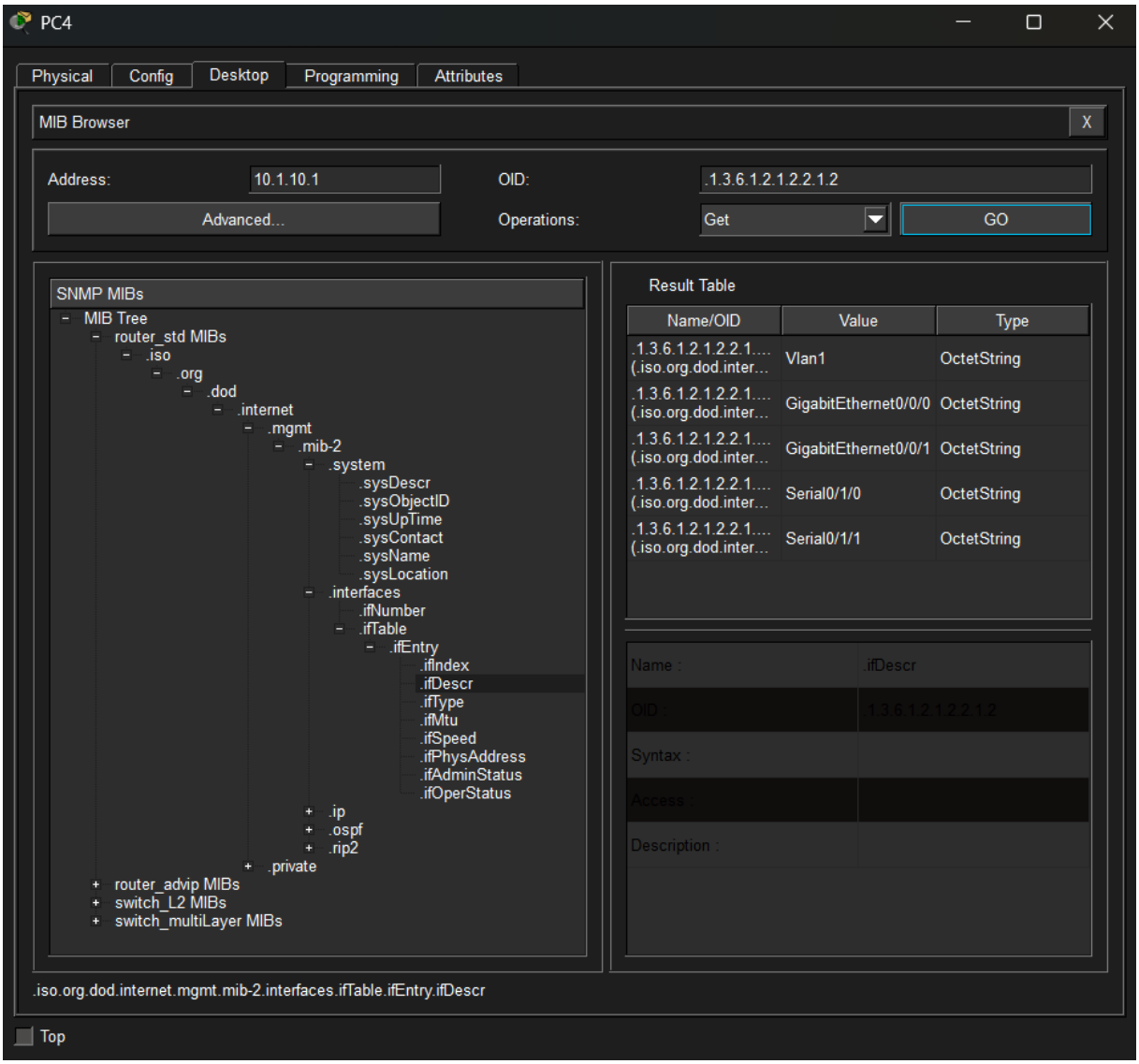


Figure 11: SNMP MIB Browser Results